

Algebraic Geometry – Exercise Sheet 2

Week 2: Prime spectra, the structure sheaf and points of affine space

Monday and Thursday afternoon sessions; Lectures 6–10

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

For a ring R , write $\gamma_R : \mathbb{Z} \rightarrow R$, $n \mapsto n1_R$, for the unique unital ring homomorphism.

Monday afternoon — material available: Lecture 6 only

Exercise 1. The affine line as a spectrum

★ CORE

Assume that $\mathbb{k}$ is algebraically closed and let $X = \text{Spec}(\mathbb{k}[t])$.

- (i) List the points of X .
- (ii) Determine which points are closed and compute the closure of each point.
- (iii) For a non-zero polynomial f , describe the closed subset $\mathbb{V}(f)$.

Exercise 2. The spectrum of the integers

★ CORE

Describe $\text{Spec}(\mathbb{Z})$: its points, its closed points, and the closure of each point. For $n \neq 0$, determine $\mathbb{V}(n)$.

Exercise 3. Nilpotents are invisible topologically

★ CORE

Let

$$A = \mathbb{k}[\varepsilon]/(\varepsilon^2), \quad X = \text{Spec}(A).$$

Determine the points and open subsets of X . Compare the underlying topological space with $\text{Spec}(\mathbb{k})$, and explain why the two affine schemes are nevertheless different.

Exercise 4. Kernels, images and their closures

★★ CORE

For each homomorphism, compute the kernel and use Proposition II.B.2 to determine the closure of the induced map on spectra.

(i)

$$\varphi : \mathbb{k}[x, y] \longrightarrow \mathbb{k}[t], \quad x \longmapsto t^2, \quad y \longmapsto t^3.$$

(ii)

$$\psi : \mathbb{k}[x, y] \longrightarrow \mathbb{k}[t, t^{-1}], \quad x \longmapsto t, \quad y \longmapsto 0.$$

Determine the image itself and exhibit explicitly the point of its closure which it misses.

(iii)

$$\chi : \mathbb{k}[x, y] \longrightarrow \mathbb{k}[t, t^{-1}], \quad x \longmapsto t, \quad y \longmapsto t^{-1}.$$

Show in this case that the image equals its closure.

Exercise 5. Closed subsets in the node

★★ CHOICE

Let $A = \mathbb{k}[x, y]/(xy)$ and $X = \text{Spec}(A)$. Denote the classes of x, y by the same letters.

- (i) Show that $\mathbb{V}(x) \cup \mathbb{V}(y) = X$ and determine $\mathbb{V}(x) \cap \mathbb{V}(y)$.
- (ii) Compute the closures of the points (x) and (y) .
- (iii) Show that neither of these two points is closed.

Thursday afternoon — material available: Lectures 6–9

Exercise 6. The affine line over several fields

★★ CORE

Let K be a field and let $X = \text{Spec}(K[u])$.

- (i) Describe all points of X , their closures, and their residue fields in terms of irreducible polynomials in $K[u]$.
- (ii) Specialise the description to $K = \mathbb{C}$, $K = \mathbb{R}$, and $K = \mathbb{F}_p$. Which closed points are K -rational?

- (iii) Take $K = \mathbb{C}(t)$. Give a K -rational closed point and a non-rational closed point, and compute their residue fields.

Exercise 7. Principal opens in the node

★★ CORE

Let $A = \mathbb{k}[x, y]/(xy)$ and $X = \text{Spec}(A)$.

- (i) Describe $D(x)$ and $D(y)$ as affine schemes.
- (ii) Show that $D(x) \cap D(y) = \emptyset$ and identify the point missing from $D(x) \cup D(y)$.
- (iii) Compute the residue fields at (x) , (y) and (x, y) .

Exercise 8. Sections and a stalk on the affine line

★ CORE

Let $X = \text{Spec}(\mathbb{k}[t])$ and $U = D(t(t-1))$.

- (i) Compute $\mathcal{O}_X(U)$.
- (ii) Compute the stalk $\mathcal{O}_{X, (t-a)}$ and its residue field.
- (iii) For which a does $(t-a)$ belong to U ?

Exercise 9. A subscheme of the affine line with two support points

★★ CORE

Let

$$Z = \text{Spec}(\mathbb{k}[t]/(t^3(t-1)^2)).$$

- (i) Determine the underlying topological space of Z .
- (ii) Decompose its coordinate ring as a product of two rings.
- (iii) Compare the two local pieces with the reduced points at 0 and 1.

Exercise 10. Prime ideals and scheme structure in the affine plane

★★ CORE

Assume that $\mathbb{k}$ is algebraically closed and work in $A = \mathbb{k}[x, y]$. Put

$$I = (xy), \quad J = (x^2y), \quad Q = (x, y)^2.$$

- (i) Using the description of the prime ideals of A , determine the primes containing I , J , and Q . Hence compare the three underlying closed subsets.
- (ii) Verify in these cases the formula

$$\sqrt{\mathfrak{a}} = \bigcap_{\mathfrak{p} \supseteq \mathfrak{a}} \mathfrak{p}.$$

- (iii) Explain how $\text{Spec}(A/I)$, $\text{Spec}(A/J)$, and $\text{Spec}(A/Q)$ retain different scheme structures even when their points coincide.

Exercise 11. Products of rings and disjoint unions

★★ CHOICE

Let B and C be non-zero rings.

- (i) Show that every prime ideal of $B \times C$ is of the form $\mathfrak{p} \times C$ or $B \times \mathfrak{q}$.
- (ii) Deduce that

$$\text{Spec}(B \times C) \simeq \text{Spec}(B) \amalg \text{Spec}(C).$$

- (iii) Relate this to the idempotents $(1, 0)$ and $(0, 1)$.

After Friday's arithmetic lecture or homework

Exercise 12. Prime ideals in $\mathbb{Z}[\sqrt{3}]$

★★ CHOICE

Let

$$A = \mathbb{Z}[s]/(s^2 - 3), \quad f = \text{Spec}(\gamma_A) : \text{Spec}(A) \longrightarrow \text{Spec}(\mathbb{Z}).$$

For a prime p , describe the points of $\text{Spec}(A)$ mapping to (p) by factoring $s^2 - 3$ over $\mathbb{F}_p$. Treat separately $p = 2$, $p = 3$, and $p \neq 2, 3$.

Exercise 13. An arithmetic conic

★★★ OPTIONAL

Put $A = \mathbb{Z}[x, y]/(x^2 + y^2 - 5)$. For

$$X = \operatorname{Spec}(A) \xrightarrow{\operatorname{Spec}(\gamma_A)} \operatorname{Spec}(\mathbb{Z}),$$

write down the equation obtained after reduction modulo a prime p . Determine for which p that equation is a square or has a singular point.